

CN MODEL PRACTICAL WRITE UP1. Extended ACL - Packet TracerAIM

To build a network in Packet Tracer using an extended Access Control List (ACL) to permit only HTTP traffic from a specific subnet to a web server while blocking all other traffic.

Procedure

no. of system : 2 PCs

Hardware

→ 1 Router

→ 1 Switch

→ 2 PCs

→ 1 Server.

Hardware IP

Device	IP Address
PC1	192.168.1.10
PC2	192.168.1.20
Server	192.168.2.10
Router G0/0	192.168.1.1
Router G0/1	192.168.2.1

Subnet Mask: 255.255.255.0

Sample steps

1. Place the router, switch, 2 PCs and server in Packet Tracer.
2. Connect PC1 and PC2 to the switch.
3. Connect the switch to the router.
4. Connect the server to the router.
5. Assign the IP addresses.
6. Configure the server as an HTTP web server.
7. Configure an extended ACL on the router.
8. Permit HTTP traffic from the required subnet.
9. Deny other traffic.
10. Test the connection using the web browser and ping.

RESULT

Thus, an extended ACL was successfully configured in packet tracer to permit HTTP traffic from the specified subnet to the web server while blocking other traffic.

2. WAN TOPOLOGY - PACKET TRACER

AIM

To design and implement a suitable WAN topology using Packet Tracer to connect 6 remote offices to a central head office and justify the selected topology.

Procedure

No. of system : 7 routers

Hardware:

- 1 Central Head Office Router
- 6 Remote office routers
- Serial / WAN links.

Sample Steps

1. Place 7 routers in packet Tracer
2. Use one router as the Central Head office router
3. Use the remaining 6 routers as Remote office routers
4. Connect each remote office router to the central router to the central router using WAN links.
5. Assign suitable IP addresses to the router interfaces
6. Configure the WAN interfaces
7. Configure routing between the offices.
8. Use the ping command to verify connectivity.

Result

Thus, a Hub-and-spoke WAN topology was successfully designed in packet Tracer to connect six remote

offices with the central head office.

3. NTP PACKET ANALYSIS - WIRESHARK

AIM

To capture and analyze NTP (Network Time Protocol) packets using Wireshark and study their header fields and payload.

Procedure

protocol : NTP

website: N/A - NTP traffic is captured from a time server.

Sample Steps

1. Open Wireshark
2. Select the active network interface
3. Start packet capture
4. Generate or wait for NTP traffic from a time server.
5. Apply the display filter.
6. Select an NTP packet
7. expand the network time protocol section
8. observe the different header fields
9. Analyze the packet information and payload.

result

Thus, NTP packets were successfully captured using Wireshark and their header fields and payload were analyzed.

4. CRC - C Programming

AIM

To implement and demonstrate cyclic Redundancy check (CRC) for a given data frame using a selected generator poly nomial.

Procedure

code: CRC implementation

file name: crc.c

Sample Steps

1. Enter the data frame
2. Enter the generator polynomial.
3. Append zeros to the data according to the generator length.
4. perform binary XOR division
5. Obtain the CRC remainder
6. Append the remainder to the original data.
7. Display the transmitted frame
8. check the received frame for errors

C Program

```
#include <stdio.h>
#include <string.h>

int main()
{
    char data[50], gen[20];
    int i, j, h, g;

    printf("Enter data: ");
    scanf("%s", data);

    printf("Enter generator: ");
    scanf("%s", gen);

    h = strlen(data);
    g = strlen(gen);

    // Add zeros
    for (i = 0; i < g - 1; i++)
    {
        if (data[i] == '1')
        {
            for (j = 0; j < g; j++)
            {
                if (data[i+j] == gen[j])
                    data[i+j] = '0';
                else
                    data[i+j] = '1';
            }
        }
    }

    printf("CRC: ");

    for (i = h; i < h+g-1; i++)
        printf("%c", data[i]);
}
```



```

return 0;
}

```

Sample Input :

Enter data: 110101101101

enter generator: 10011

Sample output:

CRC: 0000

result

Thus, the CRC was successfully implemented in C using a generator polynomial to calculate the CRC remainder and transmitted frame.

① Transport layer $\rightarrow$ Process - to - process del.
Network layer $\rightarrow$ Host - to - Host
Data link $\rightarrow$ Node to Node

② Data link $\rightarrow$ MAC add

Network $\rightarrow$ IP add

Transport $\rightarrow$ port no.

③ Data $\rightarrow$ Segment $\rightarrow$ Packet $\rightarrow$ Frame $\rightarrow$ Bits
(data link to physical)

④ Transport - Segment

Network - packet

Data link $\rightarrow$ frame

Physical $\rightarrow$ bits

⑤ SMTP $\rightarrow$ Application

TCP $\rightarrow$ Transport

IP $\rightarrow$ Network

Ethernet $\rightarrow$ Data link

⑥ Network $\rightarrow$ IP

Data link $\rightarrow$ MAC

Transport $\rightarrow$ port no.